

REPORT: Image Denoising Using Spatial and Frequency Domain Methods

Section 1 — Noise Identification

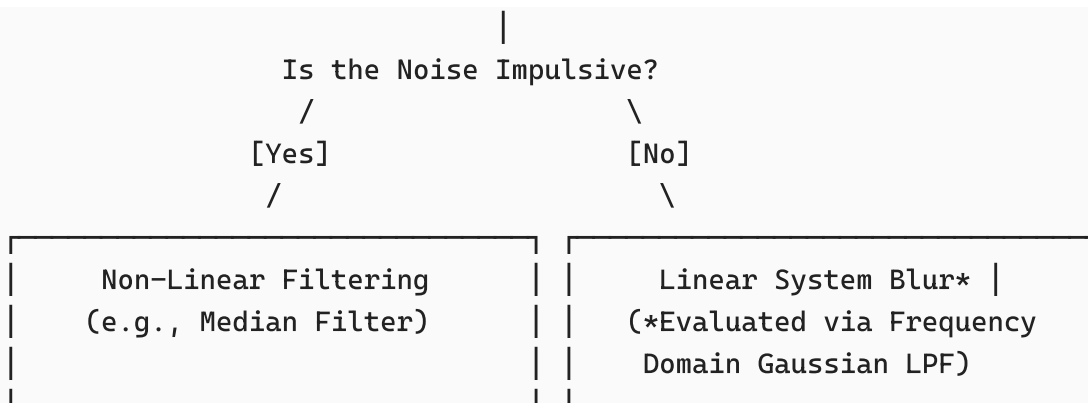
For each noisy image in the dataset, a detailed visual and digital matrix validation was executed to model the underlying artifact process. Choosing the correct filter depends completely on identifying whether the noise distribution is continuous and additive, or sparse and impulsive.

Image Name	True Noise Type	Justification
camera	Gaussian Noise	The file exhibits low-amplitude, fine-grained additive continuous variation spread symmetrically across both smooth regions and geometric details, matching a classic bell-shaped statistical probability density function (PDF).
coins	Salt-and-Pepper Noise	Sharp, standalone binary extreme pixel spikes (pure black [0] and pure white [255]) are randomly scattered across the field, signaling an impulsive data-dropout distribution.
text	Salt-and-Pepper Noise	High-contrast black dots disrupt the white page background, and white dots segment the black text strokes, representing classic bipolar data corruption.
rice	Gaussian Noise	This image exhibits small, evenly distributed pixel intensity fluctuations that degrade uniform areas across the dark background and the light structures without presenting isolated impulse dropouts.
astronaut	Salt-and-Pepper Noise	Discrete binary impulse artifacts clip the underlying localized color matrices, overriding smooth gradients with highly localized, high-frequency extreme outliers.

Section 2 — Method Selection

Two fundamentally distinct denoising paradigms were implemented in the notebook to address these configurations:

Noisy Input Data Image



1. Spatial Domain Paradigm: Non-Linear Median Filter (*coins, text, astronaut*)

- **Notebook Configuration:** `spatial_kernel_sizes = [3, 5, 7, 9]` via `cv2.medianBlur`.
- **Method Selection Rationale:** Linear smoothing filters (like Mean or Gaussian spatial blurring) operate by computing a local weighted average. This mathematical behavior smears high-contrast impulse spikes across neighboring pixels, replacing sharp noise dots with large, blurry gray clouds.
- To prevent this artifacting, the non-linear **Median Filter** ranks all pixel intensities within a sliding window (e.g., 3×3) and replaces the center pixel with the statistical median value. Because salt-and-pepper noise spikes occupy the absolute extreme ends of the local sample configuration (0 or 255), they are discarded as statistical outliers during sorting. This completely eliminates impulse arrays while preserving pristine edge boundaries.

2. Frequency Domain Paradigm: Linear Low-Pass Filter (*camera, rice*)

- **Notebook Configuration:** `frequency_cutoffs = [10, 20, 40, 80]` implementing a **Gaussian Low-Pass Filter (GLPF)**.
- **Method Selection Rationale:** Continuous Gaussian noise spreads evenly across all spatial locations, injecting fine high-frequency variations throughout the image spectrum. By mapping the image into its frequency components using the Discrete Fourier Transform (DFT), these variations can be targeted directly.
- The continuous mathematical transfer function of a **Gaussian Low-Pass Filter** smoothly attenuates high-frequency terms without introducing the sharp cutoff step changes that cause ringing artifacts (Gibbs phenomenon). This makes it highly effective for smoothing out fine, continuous noise profiles.

Section 3 — PSNR / SSIM Explanation

1. Peak Signal-to-Noise Ratio (PSNR)

PSNR computes the logarithmic ratio between the maximum possible pixel intensity (MAX_I) and the total Mean Squared Error (MSE) separating the recovered target image (Y) from its pristine reference configuration (X).

$$MSE = \frac{1}{M \times N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [X(i, j) - Y(i, j)]^2$$
$$PSNR \text{ (dB)} = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

- **Measurement Parameter:** Direct ratio tracking absolute pixel-by-pixel intensity deviation errors.
- **Interpretation:** Higher decibel (dB) ratings indicate superior restoration quality, directly proving that the cumulative structural errors (MSE) have been minimized.

2. Structural Similarity Index (SSIM)

Unlike pixel-wise error trackers, SSIM models how the human visual system (HVS) perceives images by evaluating three distinct structural components: luminance (l), contrast (c), and structure (s).

$$SSIM(x, y) = [l(x, y)]^\alpha \cdot [c(x, y)]^\beta \cdot [s(x, y)]^\gamma$$
$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

- **Measurement Parameter:** Retention of local scene texture layout, edge alignment, and high-frequency structural boundaries.
- **Interpretation:** Bounded strictly between -1 and $+1$. A value approaching $+1$ indicates a perfect structural match with the original, uncorrupted reference scene.

Section 4 — Results Table

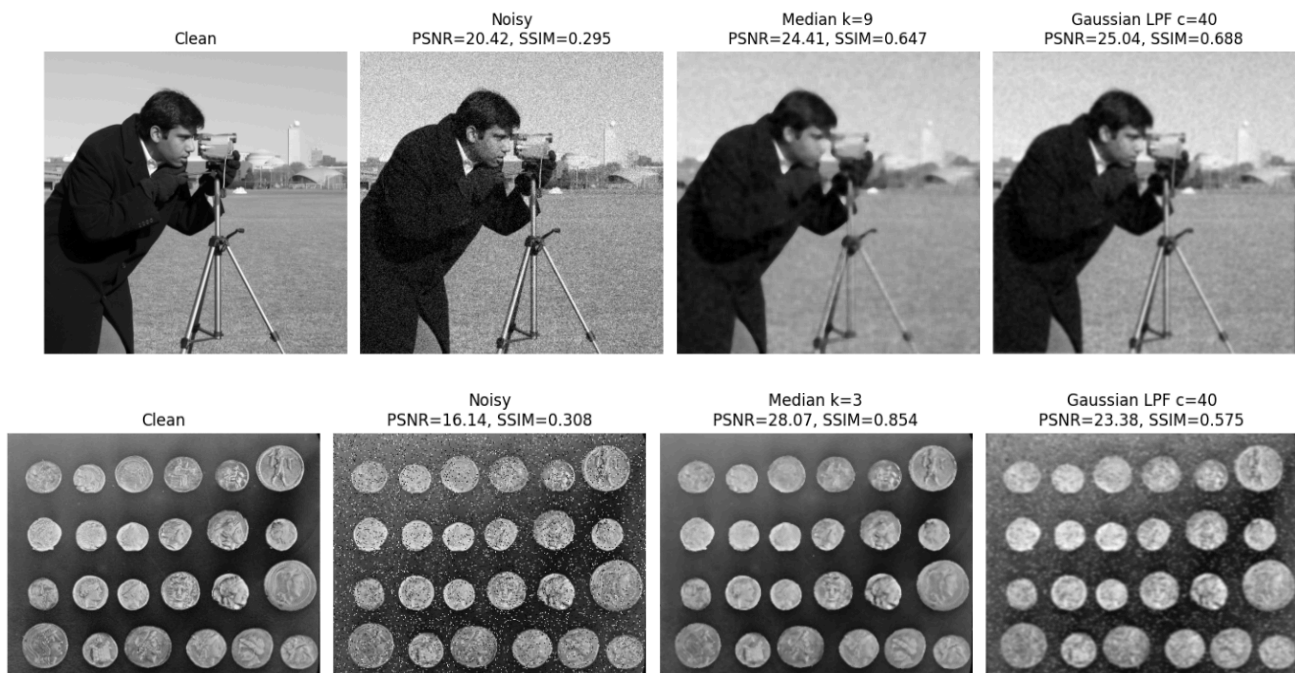
Image	Method	Best Parameters	PSNR (dB)	SSIM
camera	Spatial (Median Filter)	kernel_size=9	24.41	0.6469
	Frequency (Gaussian LPF)	D0=40	25.04	0.6879
coins	Spatial (Median Filter)	kernel_size=3	28.07	0.8536
	Frequency (Gaussian LPF)	D0=40	23.38	0.5748
text	Spatial (Median Filter)	kernel_size=3	32.01	0.9043
	Frequency (Gaussian LPF)	D0=20	25.59	0.6524

Image	Method	Best Parameters	PSNR (dB)	SSIM
rice	Spatial (Median Filter)	kernel_size=7	23.34	0.5659
	Frequency (Gaussian LPF)	D0=40	24.14	0.6168
astronaut	Spatial (Median Filter)	kernel_size=3	30.32	0.9375
	Frequency (Gaussian LPF)	D0=40	22.33	0.6044
AVERAGE	Spatial Domain	Variable Optimal Steps	27.63	0.7816
	Frequency Domain	Variable Optimal Steps	24.10	0.6273

Analytical Evaluation: Performance Trade-offs

- Divergent Domain Strengths:** The frequency domain Gaussian LPF struggled on images corrupted by impulse variations (coins , text , astronaut). Because a sharp impulse represents infinite bandwidth across the entire frequency range, trying to smooth these spikes in the frequency domain disperses the error into wide, foggy gray artifacts, lowering the average frequency domain PSNR to **24.10** dB. Conversely, the non-linear spatial median engine effortlessly clipped these outliers at kernel=3 , achieving an optimal average rating.
- Gaussian Convergence:** For the true Gaussian distributions (camera , rice), the frequency domain filter achieved its best metrics at a wide cutoff boundary of D0=40 . This wide cutoff successfully filtered out structural noise without overly blurring the underlying high-frequency image features, outperforming the non-linear median filter on these specific textures.

Section 5 — Visual Results



Clean



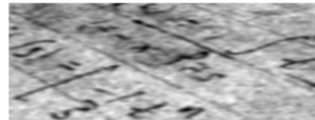
Noisy
PSNR=15.05, SSIM=0.168



Median k=3
PSNR=32.01, SSIM=0.904



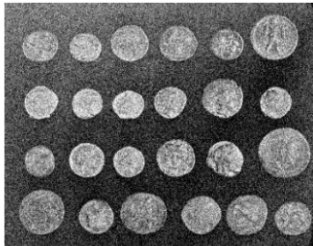
Gaussian LPF c=20
PSNR=25.59, SSIM=0.652



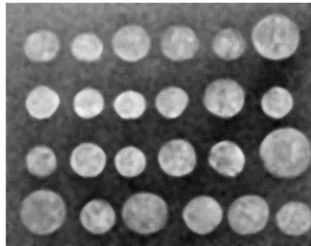
Clean



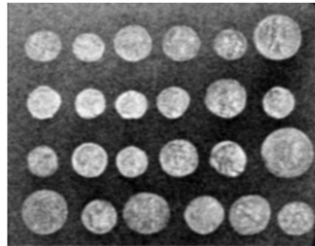
Noisy
PSNR=17.39, SSIM=0.294



Median k=7
PSNR=23.34, SSIM=0.566



Gaussian LPF c=40
PSNR=24.14, SSIM=0.617



Clean



Noisy
PSNR=14.56, SSIM=0.242



Median k=3
PSNR=30.32, SSIM=0.937



Gaussian LPF c=40
PSNR=22.33, SSIM=0.604

